



2950 Niles Road, St. Joseph, MI 49085-9659, USA
269.429.0300 fax 269.429.3852 hq@asabe.org www.asabe.org

An ASABE Meeting Presentation

Paper Number: 1111578

Adaptive Modeling for Weight Control in Vertical Auger Filling Applications of Food Powders

Cale Boriack, P.E, Project Engineer

McCormick and Company, Inc. 3300 Century Circle, Irving, TX 75062.
cale_boriack@mccormick.com

**Written for presentation at the
2011 ASABE Annual International Meeting
Sponsored by ASABE
Galt House
Louisville, Kentucky
August 7 – 10, 2011**

Abstract. *Auger fillers are used to dispense powders in food and feed packaging operations. These filling applications are met with customer and governmental requirements for weight control. A Monte Carlo simulation was performed to understand the risk of not meeting the weights and measures requirements for different amounts of giveaway and different package standard deviations. An adaptive model was created in order to decrease the variability of final weights for automated 22.7 kg (50 lb) bag single auger fillers. The model takes into account previous bulk fill weight, current bulk fill weight, and previous trim fill weight to determine new bulk and trim set points. Feed back is provided, and factors are adjusted after each bag to reduce the overall variability of final weights.*

By applying a model for weight control, the team decreased variability in filling applications for 50 lb bags of food powders. This allowed the team to fill a bag effectively on single-stage automated bag fillers at up to 6.5 bags per minute yet meet customer and government requirements for weight.

Keywords. Augers, Filling, Weight Control, Food, Powder

The authors are solely responsible for the content of this technical presentation. The technical presentation does not necessarily reflect the official position of the American Society of Agricultural and Biological Engineers (ASABE), and its printing and distribution does not constitute an endorsement of views which may be expressed. Technical presentations are not subject to the formal peer review process by ASABE editorial committees; therefore, they are not to be presented as refereed publications. Citation of this work should state that it is from an ASABE meeting paper. EXAMPLE: Author's Last Name, Initials. 2011. Title of Presentation. ASABE Paper No. 11----. St. Joseph, Mich.: ASABE. For information about securing permission to reprint or reproduce a technical presentation, please contact ASABE at rutter@asabe.org or 269-932-7004 (2950 Niles Road, St. Joseph, MI 49085-9659 USA).

Introduction

In food and feed packaging, auger fillers are often used to dispense powdered products into containers. These containers must meet customer and government requirements for weight. However, many factors influence weight control in auger dispensing. Weight control and production rate are often tradeoffs in packaging operations. Accurately dispensing product at the required rate is essential to a packaging line.

In the U.S., weights and measures are regulated by individual states, local government, as well as the federal government (Hockert and Opermann, 2011). The enforcement at the federal government is provided through the NIST division of the Department of Commerce. Net weight packages must meet 2 significant requirements: average weight and maximum allowable variation (MAV). NIST handbook 133 defines that the average quantity in a lot must be greater than the declared net weight (Butcher et al., 2011). The handbook also defines the MAV as the maximum under fill limit and specifies the limit to the number of packages that fall outside of the MAV (Butcher et al., 2011). Weights for powders and feed products fall under the Category A sampling plan. In this sampling plan, 1 package may exceed the MAV in a 3201 or greater package lot with a sample size of 48. The MAV for a 22.7kg (50lb) net weight bag is 226g (0.5lb). Product manufacturers often provide some giveaway to avoid issues resulting from scale variation and sampling errors.

Weight control for a filling operation allows a manufacturer to meet or exceed government and customer requirements for packaging weights. A filling machine dispenses product with a pattern of variability in the form of a Normal or Gaussian curve (Walton, 1966). Weight control capability can be described by the mean and standard deviation for the filled packages. In order to build the weight control capability requirements, the government requirements must be translated into mean and standard deviation. The Monte Carlo method may be used along with a specification detailing the level of risk that a manufacturer is willing to take for an underweight lot to determine the weight control capability. Engineering specifications are often written with a certain safety factor to the weight control capability.

Many factors influence the weight control of auger fillers. These factors may include product, auger speed and acceleration, head pressure, agitation, auger fill control method, and the auger design (Baird, 1996). For auger fillers, some of the factors that influence control are set at time of manufacture with the machine and accessories purchased. The auger fill control method and agitation are often parameters set at the time of manufacture for the equipment. Servomotors, stepper motors, or clutch brake systems are often used with respective controllers to meter product into its container. Agitators and augers are accessories for most vertical auger fillers. Their use may vary from product to product. Auger filler level control sensing is also specified at the time of manufacture. Speed and acceleration/deceleration for stepper and servo type auger fillers are usually set points on the operator interface terminal for the filler. Product variability is often accounted for automatically through the use of feedback control. However, this feedback control method varies widely with the setup of the machine. Checkweighers and online scales are two methods for obtaining weights for feedback control. The implementation of a particular method of weight feedback control is determined by the control requirements and packaging line setup.

Auger dosing occurs when an auger is rotated. Depending on the size of package an auger may dose once or may dose twice for each package. In large packages, an auger is often set to dose twice with a bulk fill and trim fill. For the bulk fill, the auger is set to run a certain number of revolutions or to a certain weight. The auger may stop to check weight or slow down to perform a second fill commonly named a trim fill. Depending on the speed of the bulk fill cycle and the

product variability, a weight check may occur between bulk and trim fills. Hohmann (1988) describes the dosing mass of auger fillers as shown in Equation 1 and 2.

$$V_n = \left(\left(\frac{D_1^2 - D_2^2}{4} \right) - \left(\left(\frac{D_1 - D_2}{2} \right) \bullet b \sqrt{\left(\frac{(D_1 + D_2) \bullet \pi^2}{2} + s^2 \right)} \right) \right) \quad \text{Equation 1}$$

$$m = V_n \bullet n \bullet \rho_s \quad \text{Equation 2}$$

Where:

m = mass [lb, kg]

D_1 = outer pipe diameter [in, cm]

D_2 = auger shaft diameter [in, cm]

s = pitch [in, cm]

b = auger flight thickness [in, cm]

V_n = dosing volume [in³, cm³]

n = number of auger turns (dimensionless)

ρ_s = product density [lb/in³, kg/cm³]

In large augers, dosing errors can result from product density variability and an inability of the auger to hold product in its flights. Correct selection of an auger and accessories can improve the accuracy of the dosing volume and reduce product density variations.

Dosing errors may be reduced through the use of feedback control. This feedback control may take a form of an elementary proportional-derivative (PD) loop with a single package as a unit of time. The integral term of a proportional-integral-derivative (PID) loop is removed to avoid integral windup. Equations 3 and 4 show this simple PD loop.

$$e_i = G_{prop} \bullet (m_i - ms_i) - G_{diff} (m_i - m_{i-1}) \quad \text{Equation 3}$$

$$ms_{i+1} = ms_i - e_i \quad \text{Equation 4}$$

Where:

e = error term [lb, kg]

G_{prop} = proportional gain ($0 \leq G_{prop} \leq 1$, dimensionless)

m = measured mass [lb, kg]

ms = mass set point [lb, kg]

G_{diff} = differential gain ($0 \leq G_{diff} \leq 1$, dimensionless)

Subscript i = iteration indicating current bag

The objectives of this paper are to demonstrate a method to model the risk of a lot greater than 3201 packages for 3 standard deviations and to model and implement a weight control algorithm to meet a specific set of requirements with respect to mean and standard deviation.

Materials and Methods

Accurate dispensing of product is only one aspect of quality. Automated fillers are used to accurately dispense powders on high throughput lines. The team used the model AP2400G automated filler (AccelPak, Ottawa, KS) as a test machine to dispense these powders. This filler is a single-stage automated filler with a 63.5 cm (5 in) nominal auger that is powered with a 7.5kW servomotor. The filler is capable of producing approximately 6.5 bags per minute with a total fill cycle of 5.0 seconds. The filler has an online scale system to measure the net weight of each bag. A bulk and trim dose method is used to accurately dispense product. An Allen-Bradley CompactLogix PLC (Rockwell Automation, Milwaukee, WI) controls the movement and calculations within the machine. A supervisory control system is available to log all important values to a database.

The fill cycle may be split into several steps as shown in Figure 1. Four of the steps are time based steps: perform bulk fill, bulk weight check, perform trim fill, and trim weight check. In perform bulk fill, the auger turns a specified number of revolutions to reach a target net weight of about 48 lbs. This step takes about 2.25 seconds. The bulk weight is checked after the scale settles for 1.25 seconds. The perform trim fill step adds the calculated trim amount (approximately 2 lbs) and takes about 0.25 seconds. The final bag weight is measured at the completion of the scale settle time in the trim weight check. The scale settle time is usually set to 1.25 seconds.

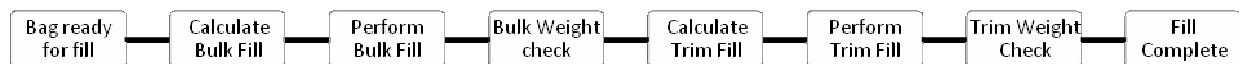


Figure 1: Fill Cycle Steps for filler.

Because of the complexity of some of the products that the team dispensed with the fillers, the team set a goal to improve the capability of the AP2400G fillers while maintaining existing rates. In improving the capability of the fillers, new quick change augers and new fill calculations were developed for the machine to reduce the variability of fill weights. Figure 2 shows a drawing of the auger shaft and the cone. The filler hopper above the removable cone is not shown. Five different augers are available for the filler for products ranging from free flowing powders to non-free flowing powders. Augers are selected based upon type of product dispensed by the machine.

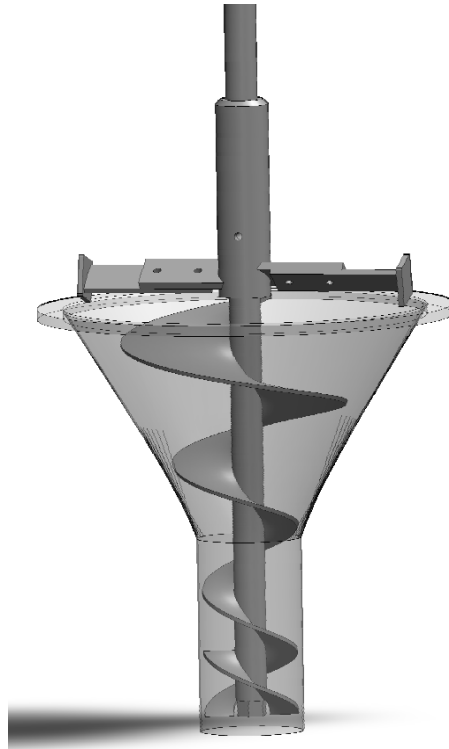


Figure 2. Filler auger and cone for AccelPak 2400G filler. The auger shown is a self feeding high compression auger for non free-flowing products.

As part of the upgrade process, the team wanted to understand the risk associated with false positives in setting the standard deviation and giveaway targets for this project. The NIST handbook 133 describes a method to determine whether a sample of package weights meets the package requirements (Butcher et al., 2011). Manufacturers use the general procedures in the handbook as a guide to determine compliance and often adapt procedures based upon the handbook to ensure compliance. A basic procedure for sampling is provided in the Figure 3.

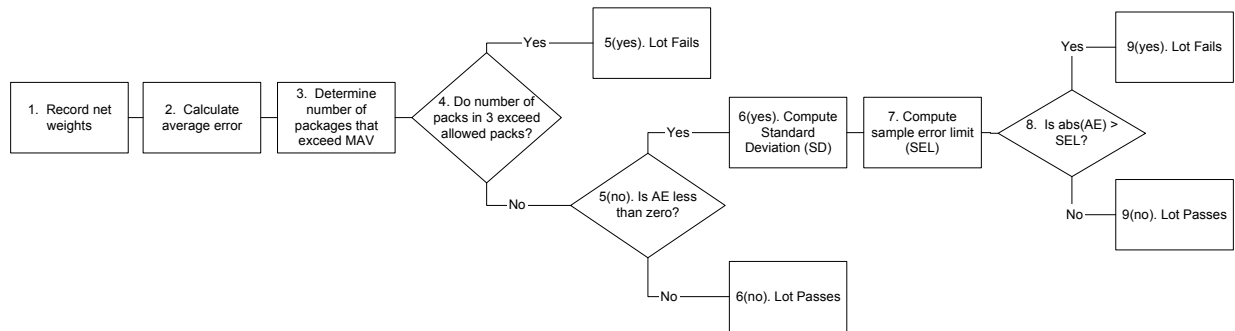


Figure 3. Basic procedure for determining if packages meet weight requirements.

In step 1, net weights are recorded for each package. The average error is calculated by subtracting the average weight from the specified package net weight. The number of underweight packages that exceed the MAV are noted in step 3. If the number of packages exceeds the allowed, then the lot fails. The average error is then compared to zero in step 5. If the average error is greater than zero, then the lot passes. If the average error is less than zero, the standard deviation is calculated. The sample error limit is calculated by multiplying the standard deviation by the sample correction factor specified in Table 2-1 of NIST Handbook 133

(Butcher et al., 2011). The absolute value of the average error is compared to the sample error limit to determine whether an underweight sample set passes.

Detailed Methods

The detailed methods consisted of the following steps:

- Estimate the risk of false positives through Monte Carlo Simulation. Monte Carlo simulation was performed on 10,000 sets of 48 samples. The inspection lot size was chosen to be more than 3200 purely to show the process. For an inspection lot size of more than 3200, the sample size is 48, the sample correction factor is 0.290, and the number of underweight packages exceeding the MAV is 1 (Butcher et al., 2011). The MAV for 22.7 kg (50 lb) net weight package is 226g (0.5 lb). Three standard deviations (0.15 lb, 0.175 lb, and 0.2 lb) were chosen to perform the simulation. Random numbers were generated in a normal distribution and arranged in a 48 x 10,000 matrix for each standard deviation. The basic procedure in Figure 3 was applied to each sample set to determine if the sample set failed. The simulation was performed using 11 values of giveaway between 0 and 0.1 lb. The number of failures were documented for each simulation and converted to a percent failure.
- Develop and log important variables such as bulk weight, bulk auger revolutions, trim weight, and trim revolutions for each bag to determine baseline data for several products. Two hundred bags for 3 products were examined to determine baseline with an offline scale set to weigh packages to 0.01 lb. Packages that were below 49.5 lb or above 50.5 lb were noted in this process.
- Determine if bulk weight has settled. If it has not, determine the best method to correct for errors in inaccurate bulk weight.
- Implement a proportional-derivative algorithm to reduce variability in the augers. Bags were weighed on the same offline scale.
- Determine set points for auger filler based upon the estimated risk from the Monte Carlo Simulation.

Results and Discussion

Results show that for a given standard deviation, the percentage of failures decrease as the giveaway increases. This occurs because the giveaway increases the number of standard deviations from the mean that the MAV is set and decreases the chance of failure from an underweight mean. Figure 4 shows the results of the simulation.

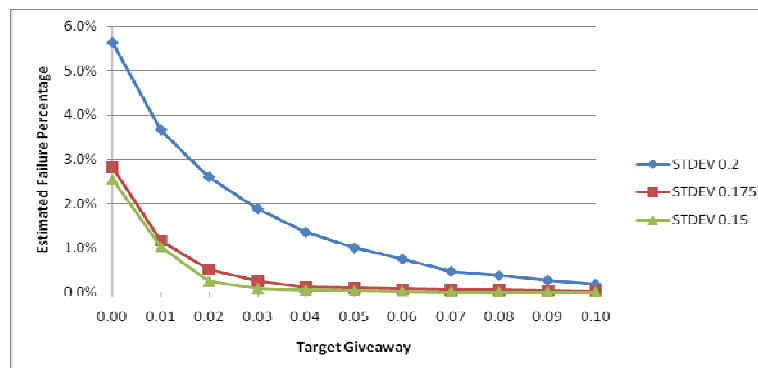


Figure 4. Estimated failure percentage for a given standard deviation and giveaway.

Once the risk was determined, a target giveaway and standard deviation target was established for the redesign of the AccelPak filler. The team worked to determine a model that could be applied in the filler to control weights. Since little was known about how the bulk density changes in the many products run on the line, a PD loop was created to adjust for the bulk and trim calculations. Because the bulk weight influences the trim weight, a study was performed to ensure that the proper bulk weight was measured. Data was collected through the trend tool for the RSLogix 5000 programming software for the PLC (Rockwell Automation, Milwaukee, WI). Figure 5 shows the data collected by the trend tool and exported to a spreadsheet for further analysis. The data showed that the bulk weight did not settle on the filler scale before the trim was calculated. The bulk weight was able to settle while the trim dose was in flight to the bag. The scale was found settled 500ms after the bulk fill weight check was considered complete. The trim fill was also found to have a flight time of about 750ms. The team did not want to reduce the rate of the machine by increasing the settle time of the scale. The bulk weight could be predicted with relatively good accuracy using 2 different methods because the scale settled before the trim influenced the weight. The first method was to perform a curve fit on the scale weights to extrapolate the settled scale value. The second method was to design a proportional loop to correct for errors between the predicted and actual settled value.

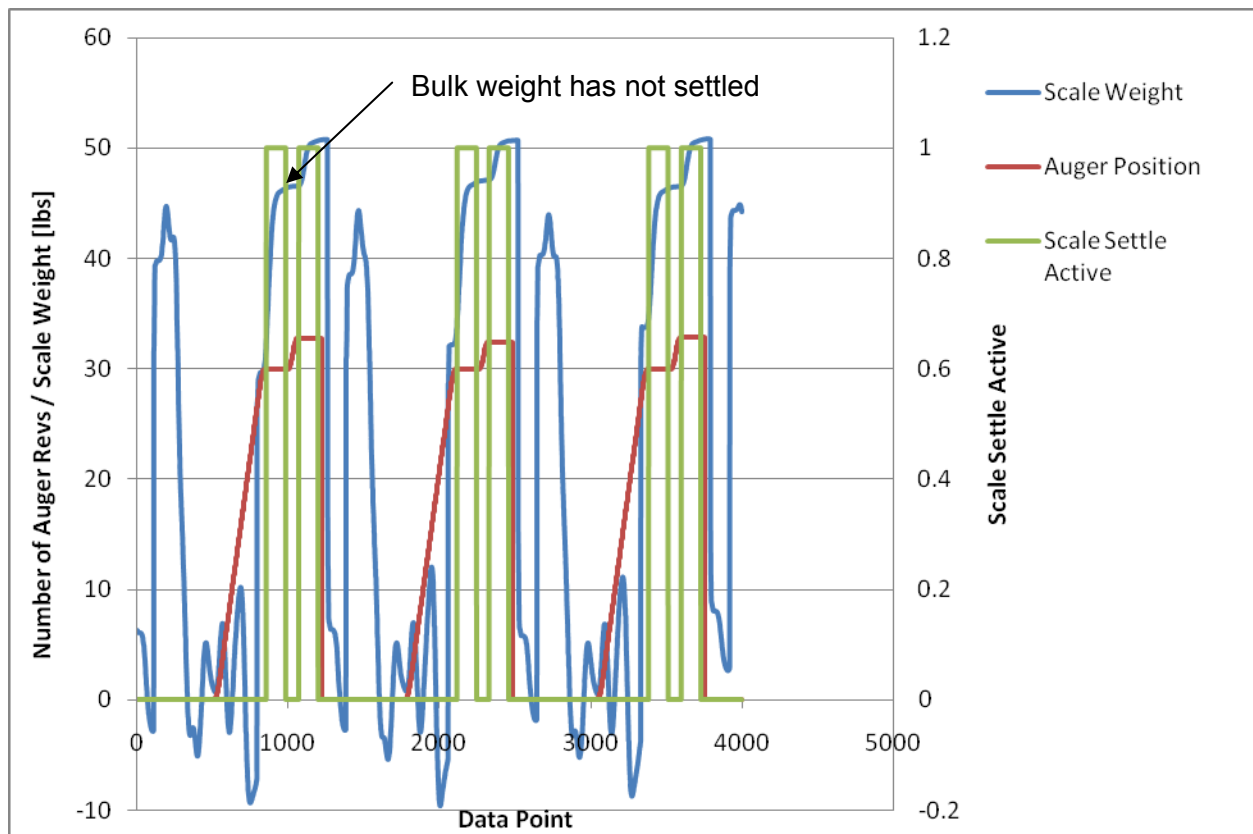


Figure 5. Weight settle analysis.

A proportional loop was created and tuned such that the approximately 1/5 of the error is corrected each bag. This results in approximately a 0.03 standard deviation of the error between the predicted weight and settled weight.

Proportional-derivative loops were created for the bulk and trim cycle to reduce the variability of final weights. The bulk loop was set with a proportional gain of 0.05 and a differential gain of

0.01. The trim loop was set with a proportional gain of 0.3 and a differential gain of 0.15. Table 1 shows the results from before and after the PD loop install.

Table 1. Results of proportional loop install.

	Product A		Product B		Product C	
Criteria	Original	New	Original	New	Original	New
Rate (packs/min)	4.9	4.9	5.2	5.0	5.8	6.2
Mean [lb]	50.09	50.03	50.04	50.05	50.15	50.06
STDEV [lb]	0.14	0.08	0.31	0.08	0.18	0.08
Overweight (packs)	1	0	17	0	22	0
Underweight (packs)	0	0	1	0	0	0

The goal for this project was to provide an average giveaway of 50.05-50.07 with a standard deviation of less than 0.15 [lb]. This would reduce the risk to less than 0.1% of a rejected lot. Using figure 4, the risk associated with a rejected lot of product B was decreased from over 1.5% to less than 0.1%. This is due to the significant improvement in weight control.

Conclusion

The team was able to reduce its risk of sending underweight product to customers through the use of PD control loops for weight control and developing an understanding of the interaction of bulk weight with trim weight. A model of risk was created to understand the interaction of giveaway and standard deviation on the risk. Significant improvements of weight control for product B decreased risks associated with shipping underweight product.

The team understands that the head pressure in the hopper plays a significant role in weight control. Work is in progress to set up the ability to determine this relationship for products before a control method is designed.

Acknowledgements

The author would like to acknowledge the engineering and operations teams for help with this project.

References

- Baird, P. 1996. Controlling filling speed and accuracy with an auger filler. *Powder and Bulk Eng.* 10(5):107-120.
- Butcher, K. S., L. D. Crown, D. A. Sefcik, and L. Warfield. 2011. Checking the net contents of packaged goods. NIST HB-133. Washington, D.C.: GPO.
- Jockert, C., and H. Oppermann. 2011. Weights and measures program requirements: A handbook for the weights and measures administrator. NIST HB-155. Washington, D.C.: GPO.
- Hohmann, H. J. 1988. Filling characteristics and dosing accuracy of volumetric dosing devices in vertical form fill seal machines. *Packaging Technol. and Sci.* 1(3): 123-135.
- Walton, D. 1966-67. Checkweighing: A review of the weight control problem and its solution. *Proc. Instn. Mech. Engrs.* 181(3E): 6-18.